INCREMENT, REGISTRY, AND MEASURE

Spatial-Temporal Co-Definition, Cyclic Return, Frequency, Scale, Numeral Bases, and Descriptive Bridges

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NOTICE - Scope discipline
Statements are tagged conceptually as Established, Constructive, Bridge/Model, or Interpretive. A bridge between two valid descriptions is not treated as proof that the descriptions are identical. Representation-dependent residue is indexed to the declared map, unit, grain, and observable.

Abstract

Measurement requires more than number. A measurable description first requires a distinction, a reference relative to which that distinction can be maintained, and an increment that can be ordered and compared. This report develops a constructive account of incrementalization: the operation by which a process, carrier, or relation becomes sufficiently partitioned to admit succession, enumeration, recurrence, and metric comparison. Spatial and temporal descriptions receive no ontological priority here. A temporal increment requires distinguishable change, while spatial separation becomes measurable only through a reference and an operation that registers difference. The two therefore function as mutually supporting conditions of measurable description.

Cyclic return supplies a particularly clear bridge. A phase can return to the same residue class while an accumulated history advances. Frequency then reads completed returns against a temporal comparator; period reads the same relation in the reciprocal orientation. The SI second provides an exact metrological example: the defining caesium-133 frequency fixes a specified recurrence count per second. A displayed hertz value therefore compresses a stack of operations: recurrence identification, cycle enumeration, temporal calibration, unit selection, and numeral representation.

The report then separates value, ratio, description, and positional representation. Base choice does not create primality or physical ratios, but it changes which divisibilities terminate, which periodicities become visually immediate, and how scale is registered by position. A duodecimal system using digits 0-9, A, B makes the factors of twelve explicit in positional subdivision and preserves zero as both identity and place marker. This becomes especially relevant when a twelve-class pitch registry, Pythagorean prime-ratio generation, twelve-tone equal-tempered closure, and conventional decimal frequency labels are compared without silently collapsing their distinct registries.

Inscribed and circumscribed regular polygons around one circle provide a second constructive calibration. The same cyclic markers generate an inner chordal witness and an outer tangent witness. Their reciprocal perimeter ratios encode comparison orientation rather than error, while powers of a common linear ratio encode dimensional grade. At triangular minimum closure the linear ratio is exactly 1:2 and the corresponding area ratio is 1:4. These results support a broader methodological claim: define the relation before naming its value, declare the registry before writing the numeral, declare the comparator before interpreting the ratio, and preserve the generating operation when translating between frames.

A final physical-application section makes explicit the bridge that motivated the geometric inquiry. The native inscribed/circumscribed n-gon ratio is first normalized between triangular minimum closure and the continuous-circle limit. That bounded closure coordinate is then compared, as a declared model bridge rather than a derivation, with the established special-relativistic factor 1/gamma = sqrt(1-beta^2). This permits longitudinal length contraction and proper-time rate to be registered on the same bounded scale, while reciprocal orientation carries dilation. The report applies the construction to local accelerator comparisons approaching the light-speed boundary and to an illustrative atomic-clock differential, while keeping the physical law, geometric registry, and translation map explicitly separate.

Contents

1. Scope, standing, and methodological discipline

This report studies the construction of measurable description rather than proposing a new physical substance called time, space, number, or frequency. Its principal question is operational: what commitments must a description introduce before an observer can distinguish an increment, place it in an order, compare it with another increment, and call the comparison a measure?

The method follows a constructive Lambda discipline developed in the companion laboratory work [4,5]. When a familiar analytic expression hides the operation that generated it, the report expands backward to a more primitive relation when possible. Finite decimals, trigonometric coordinates, logarithmic coordinates, and named transcendental constants may still serve as valid derived charts; they do not automatically serve as the defining layer. The goal is not to ban analytic mathematics. The goal is to preserve provenance.

> **CONSTRUCTIVE - Methodological rule**
> Define the relation before naming its value. Declare the whole before subdividing it. Declare the registry before writing the numeral. Declare the comparator before interpreting the ratio. Preserve the operation when translating between descriptions.

Four claim levels recur throughout the paper. Established statements include standard mathematics, the SI definition of the second, and the ISO tuning reference. Constructive statements follow from explicitly declared registries and operations. Bridge statements define maps between otherwise distinct descriptions. Interpretive statements identify structural similarities and research consequences without promoting them to physical identity.

2. Relation, distinction, and reference

A point written in isolation carries no internally available separation. Let O denote a selected reference marker. A second marker P becomes operationally meaningful when the frame distinguishes P from O. The primitive datum is therefore not the coordinate assigned to P but the relation O -> P under a declared witnessing frame.

$$O \neq P \quad \text{and} \quad R(O,P) \text{ is declared}$$

R can encode separation, adjacency, orientation, containment, identity, or another admissible relation.

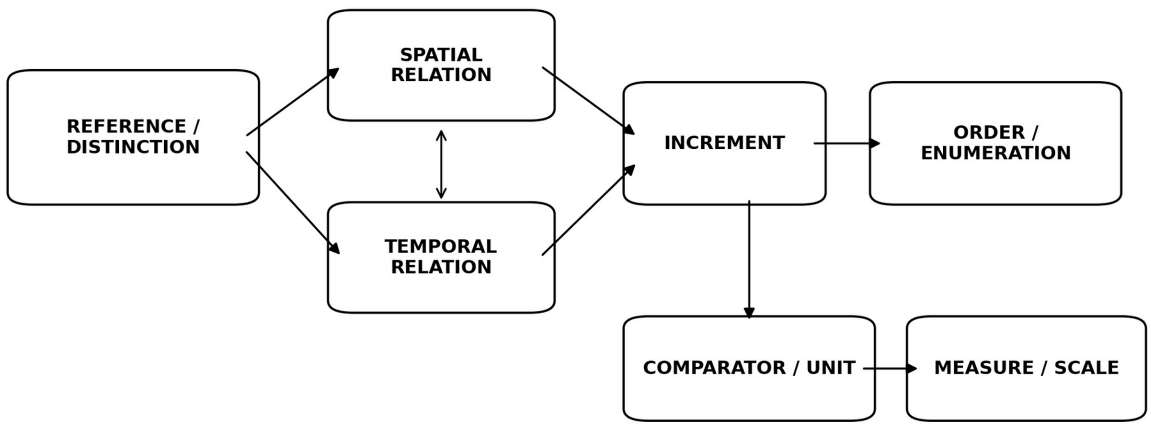
The reference does not need to be metaphysically privileged. It functions as a declared witness. Changing the witness can change coordinates, measured direction, or apparent scale without changing the underlying event record. This distinction later becomes essential for clocks: a clock reading has meaning only relative to a specified clock process, worldline, synchronization rule, and comparison operation.

> **CONSTRUCTIVE - Definition 2.1 - Distinction**
> A distinction is a frame-relative separation sufficient to prevent two selected states from being treated as the same state for the purpose of the current operation.

> **CONSTRUCTIVE - Definition 2.2 - Reference**
> A reference is a declared marker, state, process, frame, or relation against which another distinction can be compared. Reference does not imply universal privilege.

3. Spatial-temporal co-definition

The report deliberately declines to place space before time or time before space. A temporal increment requires some distinguishable change: position, state, phase, relation, registry membership, or another observable. A spatial distinction, in turn, becomes measurable only through a reference and an operation by which a separation can be registered, repeated, or compared. Each description therefore depends operationally upon structure supplied by the other.



No priority claim: spatial and temporal descriptions become measurable through declared relations, increments, and comparators.

Figure 1. Spatial and temporal descriptions enter measurement as mutually supporting relations, not as a hierarchy of ontological priority.

> **INTERPRETIVE - Co-definition principle**
> No measured time appears without a distinguishable registry of change, and no measured spatial separation appears without a reference and an operation through which difference can register. The present framework therefore treats spatial and temporal description as co-defining conditions of measurable relations.

This principle should not be confused with the stronger metaphysical statement that space and time literally generate one another. The present claim remains descriptive and operational: within a measurement account, each requires distinctions that the other helps make expressible.

4. Incrementalization, order, and enumeration

> **CONSTRUCTIVE - Definition 4.1 - Incrementalization**
> Given a carrier or process X and a frame F, incrementalization is the construction or recognition of distinguishable increments $I_0, I_1, \ldots$ sufficient to support a successor or ordering relation.

Enumeration cannot begin until something can count as one increment rather than another. Number records a distinction that the frame has already made admissible. If S denotes a successor operation, then $S(I_k)=I_{k+1}$ establishes a directed relation. An ordered registry can then attach integer markers to those increments.

$$I_0 \to I_1 \to I_2 \to \ldots \quad \text{with} \quad S(I_k) = I_{k+1}$$

Order does not yet supply metric. Knowing that I_2 follows I_1 does not tell us whether the interval corresponds to one nanosecond, one second, one meter of displacement, or one logical step in a computation. Metric enters only when the ordered registry is compared with a declared unit or calibrated process.

ESTABLISHED - Proposition 4.2 - Order before metric
An ordered sequence can be defined without assigning a metric distance between adjacent states. Any claim that adjacent registry states represent equal physical intervals therefore requires an additional calibration rule.

5. STEP, TURN, and the normalized cyclic whole

The companion First-Move Laboratory distinguishes STEP from TURN [4]. This report promotes that distinction into the measurement architecture. STEP records displacement in a selected relation; TURN records displacement within a cyclic orientation or phase registry. Neither operation, by itself, specifies meters or seconds.

$$\text{STEP: } P_k \to P_{(k+1)} \qquad \text{TURN: } q_k \to q_{(k+j \bmod n)}$$

For cyclic work, one complete turn is normalized as the whole class rather than introduced through a radian constant. The carrier can be written as the quotient T = R/Z. A rational phase k/n then represents an exact finite division of one declared whole. A radian, degree, or renderer angle can be introduced later as a chart.

$$T = R / Z, \quad [t] = [t + m] \text{ for } m \text{ in } Z, \quad \text{phase} = k/n \text{ of one full turn}$$

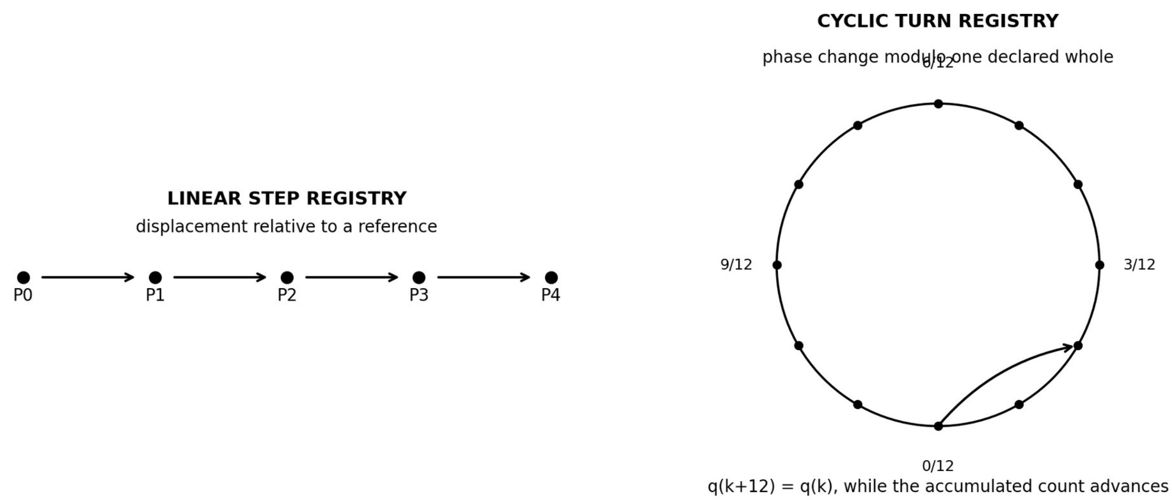


Figure 2. STEP and TURN provide complementary linear and cyclic registries. A physical unit enters only through a later bridge.

This notation prevents a convention such as 90 degrees from silently standing in for the more primitive relation one quarter of a declared full turn. The same discipline later applies to Hertz values: the physical rate and its numeral representation remain separable layers.

6. Phase return and history advance

Cyclic measurement requires two registries that ordinary notation often compresses together. The quotient registry records phase modulo a cycle. A lifted registry records how many increments have accumulated. Returning to the same phase therefore need not return to the same history.

$$q : Z \to Z/nZ, \quad q(k+n) = q(k), \quad \text{but} \quad k+n \neq k$$

ESTABLISHED - Proposition 6.1 - Phase/history separation
A cyclic projection can identify two states that differ in accumulated count. Therefore phase closure and historical return are distinct statements.

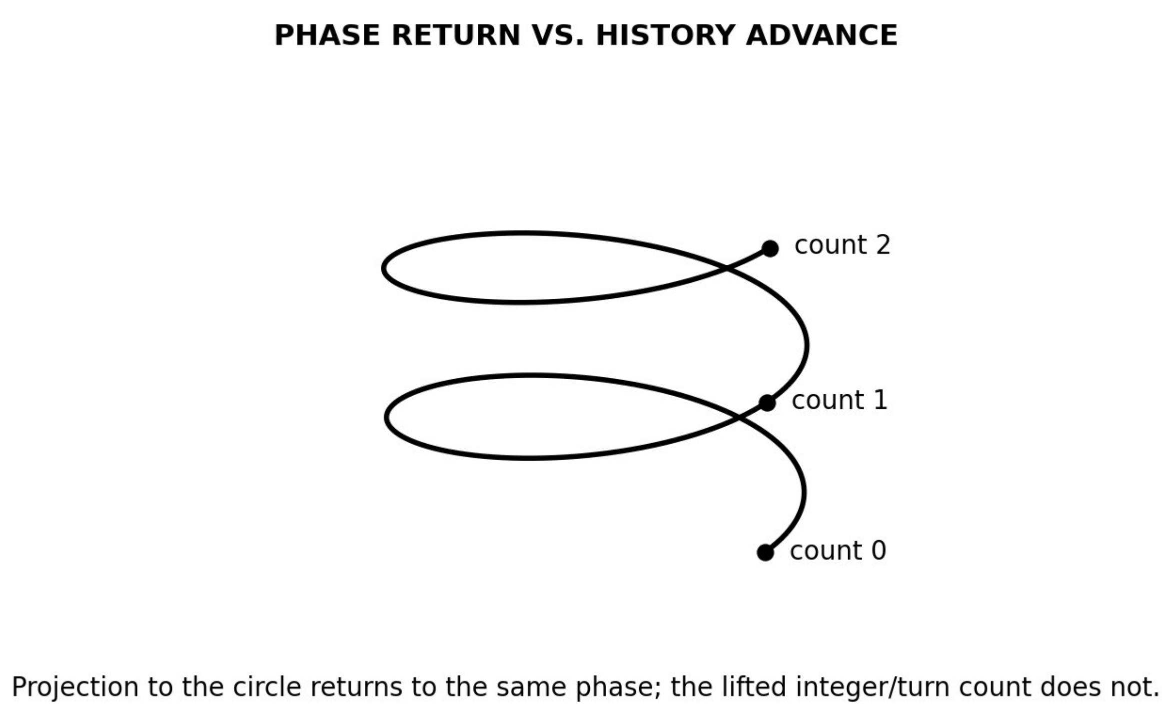


Figure 3. A circle records phase; a lifted turn count records accumulated history. Projecting the helix to the circle forgets the completed-turn index.

This proposition supplies the minimal grammar of recurrence. A second hand can return to its initial phase while the clock history advances. A vibrating string can return to an equivalent configuration while another registry records one additional oscillation. A pitch class can return modulo an octave while its exact frequency ratio has accumulated a nontrivial factor. The same mathematical pattern recurs whenever a quotient identifies return while a lift retains history.

7. Comparator, unit, measure, and bridge

A registry becomes metric only when a comparator is declared. Let k denote an enumerated turn count. A uniform temporal bridge can map that count into a coordinate time by selecting a calibrated interval Δt_F per turn.

$$\Phi_F(k) = t_0 + k * \Delta t_F$$

The bridge has three logically distinct parts: the source registry, the calibration rule, and the target quantity. Changing any one of them changes the interpretation. In a simulation, one rendered frame, one engine TURN, and one physical second need not coincide. Treating them as equal requires an explicit choice rather than a default.

CONSTRUCTIVE - Definition 7.1 - Measurement bridge
A measurement bridge Phi maps a source registry of distinctions into a target metric or descriptive registry while declaring the comparator, unit, orientation, and assumptions required for the map.

BRIDGE / MODEL - Bridge discipline
A successful bridge preserves whatever invariants its definition warrants. It does not prove that source and target descriptions are identical. Any information not transported by the bridge remains explicit residue rather than silent error.

8. The SI second as an exact recurrence calibration

The SI second provides a concrete metrological example of the architecture. The Bureau International des Poids et Mesures defines the second by fixing the numerical value of the unperturbed ground-state hyperfine transition frequency of caesium-133 to exactly 9,192,631,770 Hz, with Hz = s^-1 [1,2]. The definition therefore establishes an exact relation between a specified physical recurrence and the SI unit of time.

$$\Delta \nu_{Cs} = 9{,}192{,}631{,}770 \text{ Hz} \qquad \text{and} \qquad 1 \text{ Hz} = 1 \text{ s}^{-1}$$

In the duodecimal notation adopted later in this report, the same exact integer count is 1946716076_12. Nothing physical changes under that numeral translation. What changes is the positional representation through which the count is displayed.

$$9{,}192{,}631{,}770_{10} = 1946716076_{12}$$

This example clarifies a point that otherwise becomes easy to miss: a second does not arrive in a measurement as an unexplained primitive label. The SI defines and realizes the unit through a reproducible physical frequency standard. A practical clock then implements a bridge between local physical recurrence and an accumulated time register.

A FREQUENCY LABEL COMPRESSES A STACK OF DECLARED REGISTRIES AND BRIDGES

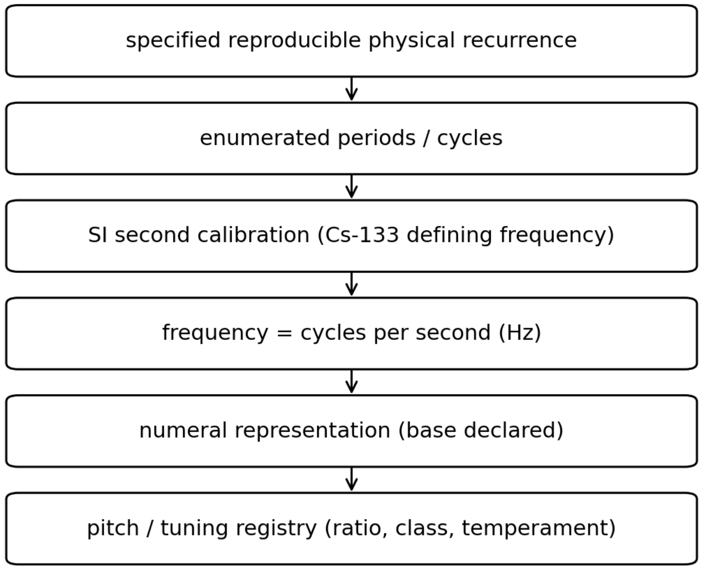


Figure 4. A frequency label compresses recurrence, counting, temporal calibration, a unit, a numeral representation, and often a second pitch or spectral registry.

9. Reciprocal orientation: frequency, period, and rate

Once a complete cycle has been defined, frequency and period read the same relation in opposite orientations. Let N_C be the number of recognized cycle completions in an interval Delta t.

f = N_C / Delta t and T = Delta t / N_C so f*T = 1

ESTABLISHED - Proposition 9.1 - Reciprocal comparison principle
For nonzero common-grade measures mu(A) and mu(B), R(A,B)=mu(A)/mu(B) and R(B,A)=1/R(A,B). The reciprocal records comparison orientation: which quantity serves as reference and which serves as referent.

A reciprocal relation should therefore not be classified automatically as measurement error. It may record an unannounced exchange of comparator. The same principle appears in length versus inverse length, wavelength versus wavenumber, frequency versus period, and the inscribed/circumscribed comparisons developed below.

10. Numeral base as a representational registry

A physical quantity, a ratio that defines it, and a numeral used to display it belong to different layers. Ordinary scientific notation often suppresses the base because decimal notation is socially default. This report restores the base marker whenever base-sensitive divisibility or scale forms part of the argument.

CONSTRUCTIVE - Distinction 10.1
Relation != description != representation. The relation specifies how quantities compare; a description selects properties relevant to a frame; a representation encodes the description in a symbol system.

This report adopts the established working convention D_12 = {0,1,2,3,4,5,6,7,8,9,A,B}. Thus A represents ten in decimal, B represents eleven in decimal, and 10_12 represents one full group of twelve. Once a calculation enters this registry, the paper keeps it there until an explicit conversion is requested.

$$D\_12 = \{0,1,2,3,4,5,6,7,8,9,A,B\}; \quad 10\_12 = 12\_10; \quad 100\_12 = 144\_10$$

The representation 440_10 Hz therefore becomes 308_12 Hz exactly, since 3*12^2 + 8 = 440. The physical rate has not changed. The display registry has.

$$440\_10 = 308\_12$$

11. The termination theorem and prime visibility

Base choice changes which rational subdivisions of a whole terminate in positional notation. This follows from a standard theorem that can be proved directly from prime factorization.

ESTABLISHED - Theorem 11.1 - Finite expansion criterion
Let a/c be a reduced rational number and let b>1 be an integer base. The base-b positional expansion of a/c terminates if and only if every prime factor of c divides b. Equivalently, c divides b^m for some finite m.

Proof sketch. If a/c terminates after m fractional places, then a/c = N/b^m for some integer N, hence c divides b^m after reduction. Conversely, if c divides b^m, then a/c can be rewritten with denominator b^m and therefore terminates after at most m places. Unique factorization supplies the prime-factor statement.

Base ten carries prime factors 2 and 5. Base twelve carries prime factors 2 and 3. Accordingly, a reduced fraction whose denominator contains only 2s and 3s terminates in base twelve. This does not make those primes more fundamental. It makes their divisibility more directly visible in that positional registry.

ratio	base 10	base 12
1/2	0.5	0.6
1/3	0.(3)	0.4
1/4	0.25	0.3
1/5	0.2	0.(2497)
1/6	0.1(6)	0.2
1/8	0.125	0.16
1/9	0.(1)	0.14
1/12	0.08(3)	0.1

The table makes the representation effect visible without changing any ratio. For example, one third repeats in decimal but terminates as 0.4_12. One fifth terminates in decimal but repeats in duodecimal. Prime structure belongs to the rational denominator; termination belongs to the relation between that denominator and the selected base.

12. Zero, position, and the Cartesian multiplication grid

Zero performs several roles that should remain distinct. It serves as additive identity; it can serve as a coordinate origin; it marks the cyclic return class in some registries; and in positional notation it preserves an unoccupied grade. The last role becomes especially visible in base twelve.

$$101\_12 = 1*12^2 + 0*12^1 + 1*12^0$$

The zero in the middle contributes no magnitude, yet removing it changes the place structure. Positional notation therefore uses zero as a witness to an empty grade, not merely as a symbol for nothing.

A multiplication table arises naturally from a Cartesian product of two digit registries. For base twelve the single-place digit set D_12 has twelve members, including zero. The multiplication field is D_12 x D_12. The zero row and zero column establish the product identity boundary and preserve the square address structure used for multiplication and division.

BASE-12 CARTESIAN MULTIPLICATION REGISTRY
Digit set: 0,1,2,3,4,5,6,7,8,9,A,B

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**A**	**B**
0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	A	B
2	0	2	4	6	8	A	10	12	14	16	18	1A
3	0	3	6	9	10	13	16	19	20	23	26	29
4	0	4	8	10	14	18	20	24	28	30	34	38
5	0	5	A	13	18	21	26	2B	34	39	42	47
6	0	6	10	16	20	26	30	36	40	46	50	56
7	0	7	12	19	24	2B	36	41	48	53	5A	65
8	0	8	14	20	28	34	40	48	54	60	68	74
9	0	9	16	23	30	39	46	53	60	69	76	83
A	0	A	18	26	34	42	50	5A	68	76	84	92
B	0	B	1A	29	38	47	56	65	74	83	92	A1

Figure 5. Base-12 multiplication as a square Cartesian registry. The zero row and column remain part of the address system, while A and B preserve one-digit values before rollover to 10_12.

CONSTRUCTIVE - Constructive reading
A square multiplication grid should not be mistaken for the primitive origin of number. It is a derived product registry formed after two independently enumerable directions have been stabilized.

13. Triangle accumulation and square product enumeration

The distinction between triangular and square enumeration provides a geometric example of how different registries organize the same act of counting. In two dimensions, three suitably joined segments provide the first polygonal enclosure. The triangle therefore represents minimum stepwise closure. The square, by contrast, arises naturally when two independently enumerable directions are combined as a Cartesian product.

Triangular accumulation: $1 + 2 + \dots + n$ | Cartesian square: $n \times n$

The familiar closed form $T_n = n(n+1)/2$ can be attached to the triangular accumulation after the construction has been made explicit. Likewise n^2 abbreviates the cardinality of the n-by-n product grid. These formulas should not erase the distinct operations that generate them: cumulative layering versus bilateral product.

CONSTRUCTIVE - Registry distinction
Triangle/simplex closure and square/Cartesian product answer different questions. One encodes minimum enclosure and cumulative layering; the other encodes independently addressable directions and multiplication.

14. Inscribed and circumscribed n-gons as finite witnesses

Let C denote a circle with center O. Choose n equally ordered markers around one full turn. Joining adjacent boundary markers generates a regular inscribed n-gon I_n. Constructing the tangent at each marker and intersecting adjacent tangents generates a regular circumscribed n-gon O_n. The same cyclic registry therefore produces an inner chordal witness and an outer tangent witness around one continuous carrier.

$$I_n \text{ subset } C \text{ subset } O_n$$

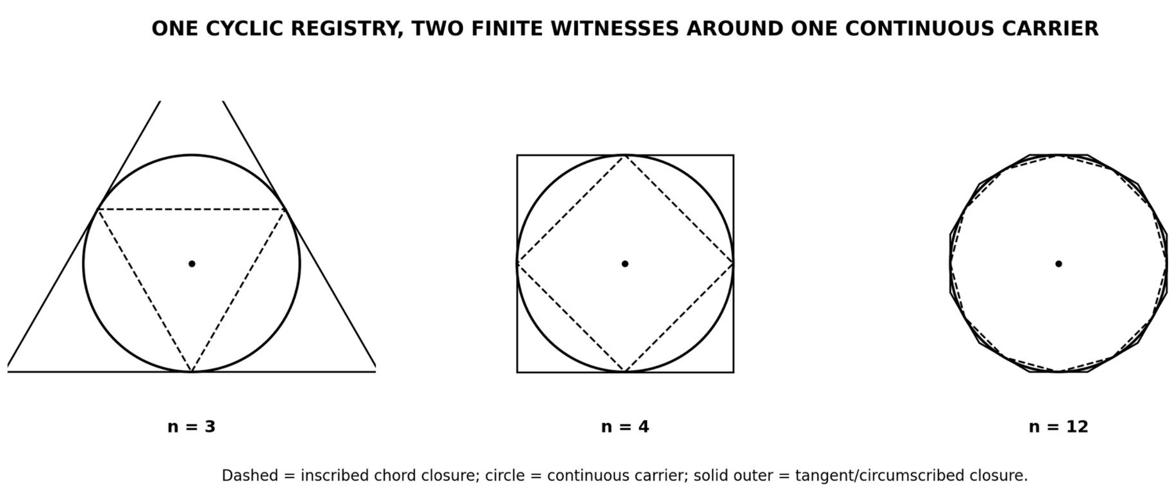


Figure 6. One finite cyclic registry generates both an inscribed and circumscribed witness around the same continuous circle. The primary construction does not require a trigonometric or pi-valued shortcut.

Let P_n^- and P_n^+ denote their perimeters. For every finite regular $n>=3$, the inscribed chordal path is shorter than the corresponding circular traversal and the circumscribed tangent path is longer. Under refinement the two finite witnesses converge toward the same circle length. A particularly clean rigorous refinement uses repeated bisection of boundary intervals: n, 2n, 4n, ... Along this sequence, inner perimeters increase, outer perimeters decrease, and the bracket narrows.

$$P_n^- < L_C < P_n^+ \quad \text{and} \quad R_n = P_n^- / P_n^+ \quad \text{with} \quad 0 < R_n < 1$$

The reciprocal P_n^+/P_n^- does not describe an error. It reverses the comparison orientation. The difference $P_n^+ - P_n^-$ can be read as a finite resolution aperture: a span left unresolved by the selected polygonal grain.

ESTABLISHED - Triangle endpoint
At n=3, the equilateral inscribed and circumscribed triangles sharing the same circle have linear scale ratio 1:2. This result follows from elementary triangle geometry without requiring sine, cosine, or a numerical value for pi.

15. Dimensional grade and repeated ratio

The triangle example clarifies why a reciprocal or multiple relation should not be called an error simply because a second measure produces a different numerical ratio. Similarity transports a linear scale factor into higher-dimensional homogeneous measures by repeated multiplication.

ESTABLISHED - Theorem 15.1 - Dimensional grade under similarity
For similar Euclidean objects related by linear scale lambda, any homogeneous d-dimensional measure scales by λ^d. Thus length scales as lambda, area as λ^2, and volume as λ^3.

For the equilateral inscribed/circumscribed pair, the linear ratio is 1:2. The area ratio is therefore 1:4. The earlier laboratory notes independently verified both values [6]. They encode the same generating scale relation at different measurement grades.

$$\text{linear: } 1:2 \quad \text{area: } (1:2)^2 = 1:4 \quad \text{volume analogue: } (1:2)^3 = 1:8$$

The constructive laboratory retains the generating ratio as provenance rather than recovering it by taking roots after the fact [4]. This is useful because the repeated-product history identifies why the higher-grade value has the form it does.

16. Vibrating strings and reciprocal scale

The present research originated in the relation among vibrating strings, prime ratios, and the ways numerical systems represent those ratios. Under the idealized comparison of strings held under equivalent wave-speed conditions, frequency varies inversely with effective vibrating length. The relation provides an exact example of reciprocal orientation between spatial extent and cyclic rate.

$$f \text{ proportional to } 1/L \quad \text{so} \quad L_1:L_2 = p:q \text{ implies } f_1:f_2 = q:p$$

The reversal should not be treated as a contradiction. Length asks how much spatial extent belongs to a cycle-supporting element. Frequency asks how many completed cycles occur per temporal unit. The numerator and comparator have exchanged roles.

ESTABLISHED - Scope note
The inverse string-length relation assumes the other parameters controlling wave speed are held fixed. The paper uses it as a calibration of reciprocal measurement, not as a universal model of every resonant instrument.

Once frequency ratios rather than absolute frequencies are compared, low-prime integer relations become especially transparent. The octave relation 2:1 and the Pythagorean fifth 3:2 can be stated without any numeral-base commitment. Base enters later when those ratios are expanded, tabulated, subdivided, or assigned positional notation.

17. Pythagorean generation, octave quotient, and exact residue

Pythagorean tuning can be modeled constructively by two exact ratio operations: multiplication by 3:2 to generate a fifth and reduction by powers of 2 to return a result to a chosen octave class. The octave therefore acts as a quotient relation, while the fifth acts as a generator.

$$\text{generator } g = 3:2 \qquad \text{octave equivalence } x \sim 2^m x$$

No finite number of exact Pythagorean fifths can equal an exact integer number of octaves. If (3/2)^m = 2^n for positive m, then unique factorization would require 3^m to equal a power of 2, which is impossible. The exact ratio lift therefore has no nonzero finite return under octave equivalence.

ESTABLISHED - Proposition 17.1 - No exact finite fifth/octave closure
There are no positive integers m,n satisfying (3/2)^m = 2^n. The result follows immediately from unique prime factorization.

Twelve fifths nevertheless approach seven octaves closely. The residual ratio is exact:

$$(3/2)^{12} / 2^7 = 3^{12} / 2^{19} = 531441 / 524288$$

This ratio is conventionally called the Pythagorean comma. The important point for the present paper is structural: two exact routes through a ratio registry approach an intended closure but do not reattach exactly. The nonzero remainder is not numerical noise. It is a mathematically exact translation residue.

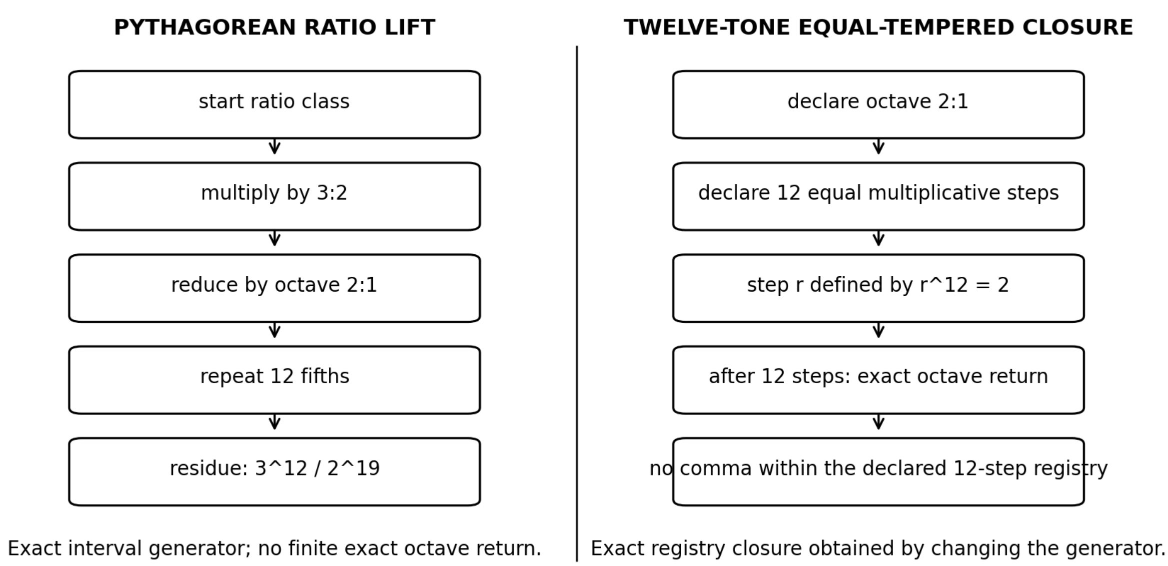


Figure 7. Exact Pythagorean ratio generation and exact twelve-tone equal-tempered closure solve different registry problems. One preserves the 3:2 generator; the other preserves exact twelve-step octave closure.

18. Twelve-class cyclic order and interval return structure

A twelve-class pitch registry has the same abstract cyclic ordering structure as a regular dodecagonal vertex registry: both can be represented by C_12 = Z/12Z. This correspondence is exact at the level of cyclic order. It does not require a claim that a polygon causes acoustic resonance.

$$C\_12 = Z / 12Z \quad \text{with} \quad \text{step } k \text{ having order } 12 / \gcd(12,k)$$

The cycle order exposes divisibility directly. A step coprime to twelve visits every class before return. A step sharing a factor with twelve closes on a proper subcycle. The perfect-fifth class step in twelve-tone pitch-class notation is 7; gcd(12,7)=1, so repeated fifth-class steps traverse all twelve pitch classes before returning to the starting class in the projected registry.

step k (base 12 digit)	common 12-TET interval name	gcd(12,k)	cycle order	return reading
0	unison	12	1	fixed class
1	minor second	1	12	full registry
2	major second	2	6	six-class cycle
3	minor third	3	4	four-class cycle
4	major third	4	3	three-class cycle
5	perfect fourth	1	12	full registry
6	tritone	6	2	two-class cycle
7	perfect fifth	1	12	full registry
8	minor sixth	4	3	three-class cycle
9	major sixth	3	4	four-class cycle
A	minor seventh	2	6	six-class cycle
B	major seventh	1	12	full registry

This table supplies an exact divisibility structure. It should not be read as a consonance ranking: cycle order measures registry return, not psychoacoustic pleasantness. Its value lies in exposing how prime factors of the chosen whole control subcycle structure.

The Pythagorean frequency lift and the twelve-class projection therefore illustrate the paper's bridge discipline particularly well. The projected class registry can return after twelve fifth-class steps, while the exact rational frequency lift retains the nontrivial comma. Projection closes; the lift records residue.

19. Twelve-tone equal temperament as declared exact closure

Twelve-tone equal temperament changes the generating relation so that a twelve-step multiplicative registry closes the octave exactly. Let r denote one equal-tempered semitone ratio. The primitive definition is the repeated-product relation

$$r * r * \ldots * r \ \ (12 \text{ factors}) = 2 \quad \text{equivalently} \quad r^{12} = 2$$

The familiar expression r = 2^(1/12) is a valid analytic abbreviation, but the constructive definition r^12=2 makes the closure condition explicit. Twelve equal steps are chosen to equal one octave exactly. The resulting fifth r^7 is therefore slightly different from the exact Pythagorean ratio 3:2.

The conventional cent system adds another registry. One octave is divided into 1200_10 equal logarithmic parts, so each twelve-tone semitone contains 100_10 cents. In the duodecimal notation of this report these counts are 840_12 and 84_12 respectively. This provides a clean example of a twelve-step pitch registry being routinely described through a decimal subdivision count.

$$1200\_10 = 840\_12 \quad \text{and} \quad 100\_10 = 84\_12$$

CONSTRUCTIVE - Representation principle
A twelve-class pitch registry, a logarithmic cent scale, and a decimal Hertz label can describe one tone simultaneously. Their coexistence does not make them the same registry. A rigorous account declares each layer and the bridge between them.

20. Decimal and duodecimal translation of standard frequencies

ISO 16 specifies 440 Hz as the standard tuning frequency for the note A in the treble stave [3]. Using A4 = 440 Hz as the anchor and r^12=2 for twelve-tone equal temperament, neighboring frequencies satisfy f_k = 440 r^k, where k counts semitone steps from A4. The table below prints the same physical frequency in the conventional decimal numeral registry and in the duodecimal registry adopted here.

$$f\_k = 440 * r\^k, \quad r\^12 = 2, \quad k \text{ in } Z$$

note	k from A4	frequency in Hz (base 10)	same Hz value (base 12, 5 fractional places)
C4	-9	261.625565	199.760B9
C#4 / Db4	-8	277.182631	1B1.22370
D4	-7	293.664768	205.7B888
D#4 / Eb4	-6	311.126984	21B.16352
E4	-5	329.627557	235.76450
F4	-4	349.228231	251.28A47
F#4 / Gb4	-3	369.994423	269.BB244
G4	-2	391.995436	287.BB414
G#4 / Ab4	-1	415.304698	2A7.37A63
A4	0	440.000000	308.00000
A#4 / Bb4	1	466.163762	32A.1B6B9
B4	2	493.883301	351.A7242
C5	3	523.251131	377.301B5

The duodecimal column is a numeral translation of the same Hertz values, not a different tuning law. Most equal-tempered frequencies remain irrational relative to the 440-Hz anchor and therefore require finite approximations in either positional base. The anchor itself converts exactly: 440_10 Hz = 308_12 Hz. The table deliberately stops mixing bases inside individual calculations; conversion appears only at the comparison surface.

CONSTRUCTIVE - Standard-value translation
When a standard frequency table must communicate with ordinary engineering or musical practice, decimal values should be given alongside the duodecimal representation. Within a base-sensitive derivation, however, one base should remain active until an explicit bridge is invoked.

20.1 Contemporary "Solfeggio frequency" lists as a base-sensitivity control

A contemporary alternative/New Age audio literature commonly circulates a six-value set (396, 417, 528, 639, 741, and 852 Hz), often extended with 174, 285, and 963 Hz to a nine-value list [9]. The present report includes that modern list only as a curiosity and a representational control. It does not treat the list as a historically established solfege tuning, and it does not endorse any biological, therapeutic, spiritual, or otherwise exceptional property attributed to the individual frequencies.

SCOPE NOTE - These values are reproduced only as a contemporary frequency list and as a test of numeral-registry translation. No special efficacy or privileged physical status is inferred from the labels.

As in the standard-frequency table above, the conversion below changes only the positional numeral used to write the same SI frequency. It does not retune the tone. A genuinely duodecimal derivation would have to declare the generating whole, the subdivision or ratio law, the increment sequence, closure/equivalence rules, and the bridge by which the result is finally reported in SI hertz. Merely rewriting a decimal hertz label in base 12 performs none of those steps.

circulated frequency label (base 10 Hz)	same integer frequency (base 12 Hz)	list position
174_10 Hz	126_12 Hz	extended lower value
285_10 Hz	1B9_12 Hz	extended lower value
396_10 Hz	290_12 Hz	commonly cited core six
417_10 Hz	2A9_12 Hz	commonly cited core six
528_10 Hz	380_12 Hz	commonly cited core six
639_10 Hz	453_12 Hz	commonly cited core six
741_10 Hz	519_12 Hz	commonly cited core six
852_10 Hz	5B0_12 Hz	commonly cited core six
963_10 Hz	683_12 Hz	extended upper value

For example, 528_10 Hz = 380_12 Hz exactly. The equality states one physical rate - 528 cycles per SI second - through two positional numeral registries. It should not be read as a new frequency, a new temperament, or a derivation of the listed value from base twelve.

The list also provides a useful control on numeral-pattern arguments. The often-emphasized decimal digit reductions of these labels belong first to the chosen representation. In base 10, summing digits preserves the represented integer modulo 9 because 10 is congruent to 1 modulo 9; in base 12, the analogous unweighted digit-sum shortcut preserves residue modulo 11 because 12 is congruent to 1 modulo 11. The integer residue modulo 9 can of course be computed in any base if modulo 9 is explicitly declared, but the familiar visual 3/6/9 digit pattern is a decimal presentation of that chosen modular relation, not by itself an acoustic invariant. No physical significance follows from the digit pattern without an independently justified bridge from the arithmetic observable to a physical one.

CONSTRUCTIVE - Tuning distinction. A finite list of absolute hertz values is not yet a tuning system. A tuning description additionally requires a reference, interval/generator relations, an equivalence or closure rule, and a mapping from those relations to pitch classes or sounding events.

21. Temporal order, causal order, and observer-local clocks

Temporal order and causal order should remain distinct. A relation A precedes B under a selected temporal ordering does not by itself establish that A caused B. A causal model adds admissibility rules such as propagation, locality, interaction channels, or other domain-specific constraints.

$$A <_t B \quad \text{does not imply} \quad A \to_{cause} B$$

This distinction becomes practical when multiple clocks are compared. Each clock accumulates its own registered history and may require a bridge to a shared coordinate description. A simulation TURN count therefore should not be identified silently with a physical proper time or coordinate time. Instead, a declared map connects engine enumeration to the chosen physical model.

$$\text{TURN } k \to t_{ref}(k) \to \tau_i(k)$$

The present paper does not derive special or general relativity from polygonal or tuning registries. Relativistic clock comparison enters as a later physical application of the bridge architecture: one must declare the reference coordinate, worldline, physical model, and observable before translating between clock readings. The same methodological rule applies: a structural analogy does not replace the governing physical equations.

BRIDGE / MODEL - Bridge boundary
The inscribed/circumscribed ratios, Pythagorean closure residue, and cyclic-return algebra provide exact mathematical calibrations of registry behavior. Any mapping from those structures to relativistic dilation, horizon behavior, or another physical effect remains a separate bridge that requires an independently justified physical law.

22. Polygonal closure as a bounded relativistic comparison bridge

Sections 14 and 15 deliberately stopped at the mathematical boundary between the polygonal registry and physical interpretation. This section supplies the deferred bridge that motivated the investigation. It does not derive special relativity from n-gons, nor does it claim that a polygon causes relativistic dilation or contraction. Instead, it asks a narrower and testable representational question: can the finite closure family provide a bounded registry on which the already-established Lorentz contraction/proper-time factor can be carried without erasing the geometric endpoints, reciprocal orientation, or the distinction between discrete construction and continuous physical state?

BRIDGE / MODEL - Standing of Section 22. The polygonal quantities below are constructed geometrically. The Lorentz relations are established special relativity. The identification between the normalized closure coordinate and 1/gamma is an explicitly declared representation bridge. Its usefulness can be tested; it is not presented as a derivation of relativity.

22.1 Native closure ratio and the minimum-closure endpoint

Reuse the construction of Section 14. For each regular n-gon with integer $n \geq 3$, let P_n^- denote the perimeter of the inscribed chordal witness and P_n^+ the perimeter of the circumscribed tangent witness generated from the same n equally ordered markers on one circle. Define the native finite comparison $R_n = P_n^- / P_n^+$. The construction gives $0 < R_n < 1$ at every finite n, $R_3 = 1/2$ at triangular minimum closure, and $R_n \to 1$ as the finite witnesses refine toward the continuous carrier.

$$R_n = P_n^- / P_n^+ \qquad R_3 = 1/2 \qquad \text{and} \qquad \lim_{n\to\infty} R_n = 1$$

The endpoint n=3 has structural status independent of any physical analogy: three straight segments provide the first ordinary polygonal enclosure in the plane. Below that value the selected object class no longer remains a closed polygon. The continuous-circle limit occupies the opposite end of the same family. The interval [1/2,1) is therefore not an arbitrary numerical crop; it is the native range of this particular inner/outer regular-polygon comparator.

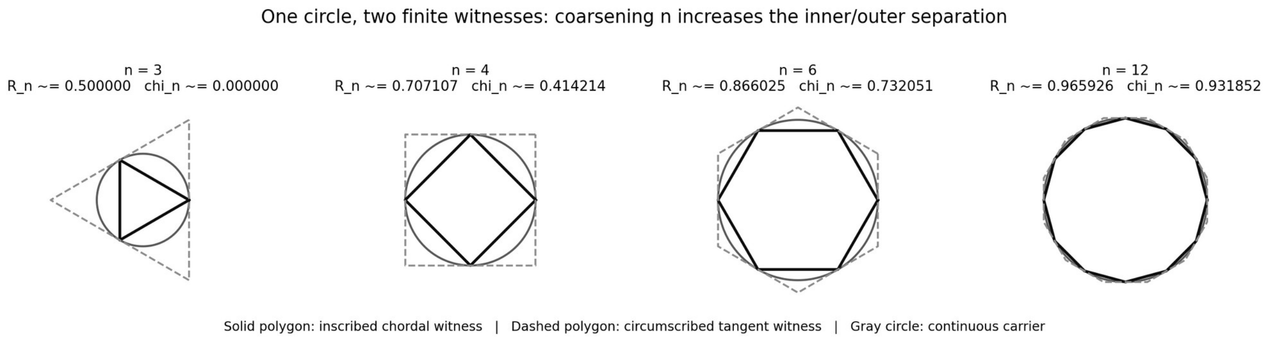


Figure 8. Native geometric family used by the bridge. The renderer uses ordinary analytic coordinates only to draw the objects; the defining data remain one circle, n full-turn markers, adjacent chords, adjacent tangents, and the measured ratio P_n^-/P_n^+.

22.2 Endpoint-preserving normalization: the closure coordinate

To compare this native interval with a physical factor whose natural range is [0,1], normalize the geometric ratio by the unique affine map that sends the triangle endpoint R_3=1/2 to 0 and the continuous limit R=1 to 1. Define the closure coordinate chi_n by

```
chi_n = (R_n - R_3)/(1 - R_3) = 2 R_n - 1
```

Thus chi_3 = 0 and chi_n -> 1 under refinement. Equivalently, the truncation aperture a_n = 1 - chi_n increases as the polygonal description is coarsened and vanishes at the continuous-circle limit. This normalization adds no physical content. It only re-expresses the exact bounded geometric family on a unit interval while preserving both endpoints and order.

CONSTRUCTIVE - Proposition 22.1 - Unique affine endpoint normalization. Among affine maps chi = a R + b, exactly one sends R=1/2 to chi=0 and R=1 to chi=1: chi=2R-1. Any nonlinear remapping would introduce additional model structure and must be declared separately.

22.3 Bridge to the Lorentz factor

For inertial relative speed v, special relativity defines beta = v/c and gamma = 1/sqrt(1-beta^2). The reciprocal factor 1/gamma = sqrt(1-beta^2) lies in (0,1] for a massive object and approaches zero as beta approaches the light-speed boundary. The same factor governs two standard comparisons aligned with the relative motion: longitudinal length contraction L_parallel/L_0 and the proper-time rate d tau/d t of the moving clock relative to the chosen inertial coordinate time [7,8].

```
beta = v/c      gamma = 1/sqrt(1-beta^2)      L_parallel/L_0 = d tau/d t = 1/gamma = sqrt(1-beta^2)
```

The proposed bridge now makes one explicit identification of coordinates: chi is used to register the established factor 1/gamma. Under this bridge, the geometric continuum limit chi->1 corresponds to beta->0, while triangular minimum closure chi->0 corresponds to the limiting beta->1 boundary. The reciprocal 1/chi registers the dilation orientation gamma. Thus length contraction versus reciprocal spatial dilation, and proper-time slowing versus coordinate-time dilation, become opposite readings of the same transported factor.

Phi_SR: chi <-> 1/gamma so beta = sqrt(1-chi^2), gamma = 1/chi

ESTABLISHED + BRIDGE - Important boundary. Special relativity supplies the Lorentz equations; the polygonal construction does not. The triangle endpoint is mapped to the kinematic light-speed boundary by the declared affine bridge. It should not be called a gravitational event horizon, and for a massive particle it is a limiting boundary rather than an attainable state.

polygon anchor n	native R_n	closure chi_n	bridged beta	bridged gamma
3	0.500000	0.000000	1 (limit)	diverges
4	0.707107	0.414214	0.910180	2.414214
5	0.809017	0.618034	0.786151	1.618034
6	0.866025	0.732051	0.681250	1.366025
8	0.923880	0.847759	0.530382	1.179580
12	0.965926	0.931852	0.362839	1.073132
24	0.991445	0.982890	0.184195	1.017408
infinity	1.000000	1.000000	0.000000	1.000000

Table 5. Discrete polygon anchors under the declared SR bridge. Decimal values use the conventional analytic chart only for numerical verification; the geometric construction remains the defining layer.

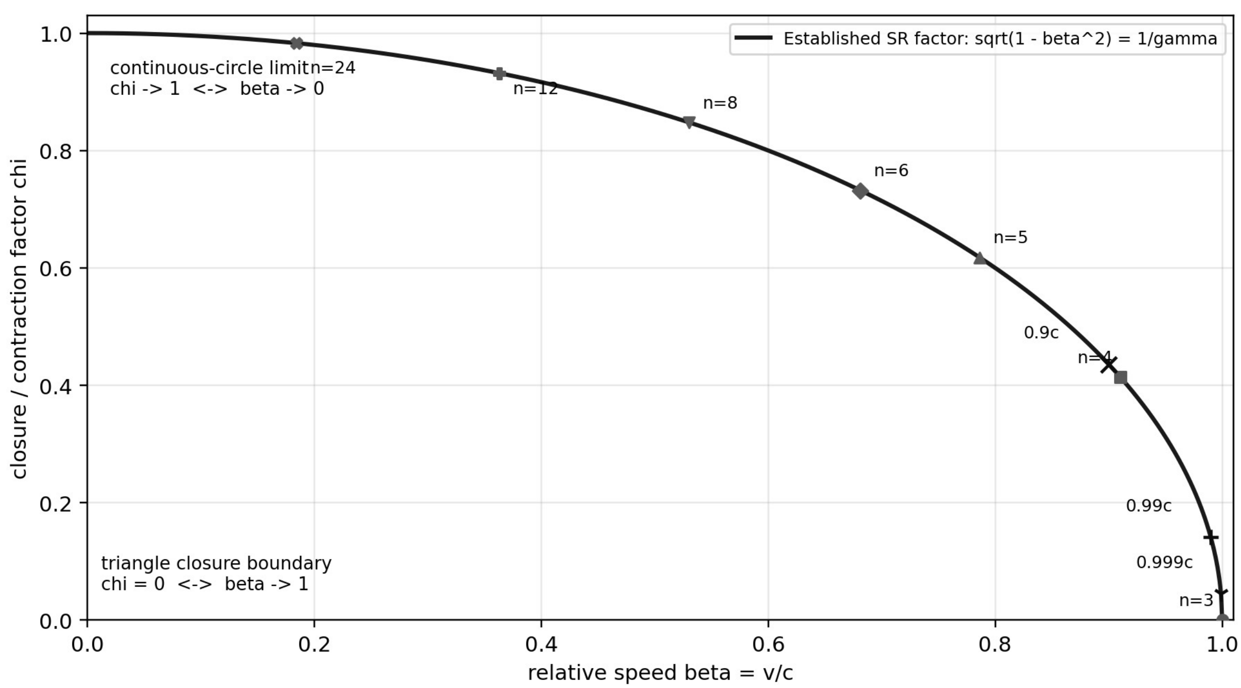


Figure 9. The established special-relativistic factor 1/gamma shown as a continuous physical curve, with discrete regular-polygon closure anchors superimposed through Phi_SR. The dots do not quantize speed; they provide finite registry markers on a continuous physical relation.

22.4 Discrete polygon anchors do not quantize continuous velocity

The side count n is an integer construction parameter, whereas relative speed beta is continuous. The bridge therefore should not manufacture a literal fractional-sided polygon for every velocity. A continuous physical state can instead be bracketed by adjacent constructed anchors. If chi_n <= sqrt(1-beta^2) < chi_(n+1), the state lies between the n- and (n+1)-gon closure markers. A continuous effective-n formula may be used as a secondary analytic chart, but it is not needed for the registry itself.

This distinction is especially important near the light-speed boundary. Many physically different values of beta close to one fall between the triangle and square anchors. Their physical differences remain fully resolved by beta, gamma, proper time, and length; the coarse polygonal anchor simply reports that all occupy the most strongly truncated region of the chosen finite closure registry.

22.5 Accelerator comparison: longitudinal contraction approaching c

Consider a local straight segment aligned with the motion of a highly relativistic particle or bunch. If L_0 is the segment length in its rest frame, an inertial frame in which it moves at speed beta c measures the longitudinal length L = chi L_0 with chi=sqrt(1-beta^2). Equivalently, reversing the comparator gives the spatial dilation factor L_0/L = gamma. For a circular accelerator one must apply this statement locally or to a specified inertial comparison; a single global comoving inertial frame around the entire accelerated ring is not assumed.

beta=v/c	L/L_0 = d tau/dt = chi	gamma = 1/chi	polygon anchor band
0.5000	0.866025	1.154701	n=8 to 9
0.9000	0.435890	2.294157	n=4 to 5
0.9900	0.141067	7.088812	n=3 to 4
0.9990	0.044710	22.366272	n=3 to 4
0.9999	0.014142	70.712446	n=3 to 4

Table 6. Accelerator-scale examples. As beta approaches one, the established contraction/proper-time factor approaches zero and the bridge moves toward the triangular minimum-closure boundary.

The geometric reading is therefore a truncation registry: increasing relative speed moves the transported factor away from the continuous-circle endpoint and toward minimum polygonal closure. The physical law remains the Lorentz transformation. The n-gon family supplies a bounded comparative picture of how much of the unit closure coordinate survives in the selected frame.

22.6 Clock differentials as oriented reciprocal closure readings

Clock comparison adds a useful extension because a relative clock ratio may lie on either side of one when velocity and gravitational potential are both present. Let q_i = Delta tau_i / Delta tau_ref over one declared comparison interval. To register the magnitude on the bounded closure coordinate without losing which clock leads, define an oriented pair

```
chi_clock = min(q_i, 1/q_i)      and      sigma_i = sign(q_i - 1)
```

The bounded factor chi_clock records the magnitude of the reciprocal departure from unity; sigma_i records orientation. A slower clock has q_i<1 and negative orientation. A faster clock has q_i>1 and positive orientation. This is a representation rule for clock differentials, not an additional law of clock dynamics.

As a concrete calibration, use the illustrative weak-field aircraft comparison developed in the companion studio: duration T=41 h, altitude h=10,000 m, airspeed u=250 m/s, representative latitude 40 degrees, Earth rotation included through v_g = Omega R cos(phi), and the weak-field/low-speed approximation

```
Delta tau_(plane-ground) ~= T [ g h/c^2 - (v_plane^2 - v_g^2)/(2 c^2) ],      v_east=v_g+u,      v_west=v_g-u
```

With the declared parameters this model gives approximately -36.55 ns for the eastbound clock relative to ground and +256.01 ns for the westbound clock relative to ground after 41 h. These are parameterized model outputs, not hard-coded historical Hafele-Keating observations. NIST documents the experimentally established facts that relative velocity and gravitational potential produce measurable clock-rate differences [7].

clock comparison	Delta tau (ns)	q = tau_i/tau_ref	bounded chi_clock	orientation
ground reference	+0.00	1.000000000000000	1.000000000000000	reference
eastbound - ground	-36.55	0.999999999999752	0.999999999999752	slower (-)
westbound - ground	+256.01	1.000000000001734	0.999999999998266	faster (+)

Table 7. Illustrative aircraft clock differential registered as a bounded reciprocal magnitude plus an orientation sign. Both differences lie extremely close to the continuous-circle endpoint at ordinary geometric display scale.

The result clarifies why reciprocal structure matters. If only chi_clock were retained, the eastbound and westbound cases would both appear as tiny departures from unity and the direction of the clock difference would be lost. The pair (chi_clock, sigma) preserves both magnitude and orientation. This is the temporal counterpart of keeping P_n^-/P_n^+ distinct from its reciprocal rather than treating one as an error.

22.7 What the bridge establishes, and what remains open

EXACT / ESTABLISHED - The regular-polygon family has a minimum closure n=3; R_3=1/2; R_n tends to 1 under refinement; chi=2R-1 is the unique affine endpoint normalization. Special relativity independently establishes 1/gamma=sqrt(1-beta^2), longitudinal length contraction, and proper-time dilation for inertial comparisons.

BRIDGE / MODEL - Phi_SR identifies the normalized closure coordinate chi with the established factor 1/gamma so that the two endpoint structures can be compared on one bounded registry. Clock ratios outside the pure-SR one-sided case are represented by reciprocal magnitude plus orientation.

NOT CLAIMED - The paper does not claim that n-gons generate Lorentz symmetry, that c can be derived from triangle geometry, that velocity is physically quantized by polygon side count, that the triangle is a gravitational event horizon, or that the aircraft approximation replaces a full relativistic worldline integration. The bridge remains valuable only to the extent that it preserves distinctions and clarifies comparisons without smuggling in those stronger claims.

The bridge is therefore deliberately modest but nontrivial. The original geometric observation - curve truncation by successively coarser polygonal closure, bounded below by the triangle - now receives an explicit place in the measurement architecture. It can register both spatial and temporal Lorentz factors, it retains reciprocal dilation as orientation rather than contradiction, and it provides finite closure anchors against a continuous physical curve. Whether this structural correspondence supports deeper physics remains an open research question; the present result is the formalization of the correspondence itself.

23. Claims, limitations, and falsification conditions

The report makes several claims that can be checked independently of its broader interpretation. The following list separates what the construction establishes from what remains a research proposal.

- Exact: cyclic step order in C_n is n/gcd(n,k); prime n yields full period for every nonzero step.
- Exact: a reduced rational terminates in base b exactly when its denominator divides some power of b.

- Exact: the equilateral inscribed/circumscribed triangle pair sharing one circle has linear ratio 1:2 and area ratio 1:4.
- Exact: no positive integers m,n satisfy (3/2)^m = 2^n; twelve fifths versus seven octaves leave exact ratio 531441/524288.
- Exact by definition: twelve-tone equal temperament selects r such that r^12=2.
- Established metrology: the SI second is defined using the fixed caesium-133 defining frequency; Hz equals s^-1.
- Established standard: ISO 16 specifies 440 Hz as standard tuning frequency for A.
- Constructive: STEP, TURN, incrementalization, registry, bridge, and resolution aperture are operational terms introduced or consolidated for this research program.
- Interpretive: reciprocal relations can often be read profitably as oriented comparisons; multiple ratios across grade can encode repeated scale rather than contradiction.
- Open bridge: no physical theorem is claimed that identifies polygonal closure limits with relativistic horizons or derives physical dynamics from numeral base.
- Scope curiosity: the modern "Solfeggio frequency" list in Section 20.1 is included only as a base-translation and representation control; no therapeutic, biological, spiritual, or privileged acoustic property is claimed.

The framework would be weakened by any example in which a claimed exact relation depends upon an undeclared coordinate shortcut, any base-sensitive conclusion that disappears when the base is made explicit, or any physical interpretation that cannot state the governing law connecting the mathematical registry to the physical observable. These are useful failure conditions rather than rhetorical exceptions.

- Established physics: for inertial comparisons, the Lorentz factor satisfies gamma=1/sqrt(1-beta^2), longitudinal length contracts by 1/gamma, and moving proper time accumulates at rate d tau/dt=1/gamma.
- Bridge/model: chi=2(P_n^-/P_n^+)-1 is the unique affine normalization of the polygonal closure interval, and Phi_SR: chi <-> 1/gamma is a declared representational bridge. It is not a derivation of special relativity or a physical quantization of velocity.

24. Conclusions

The central result of this report is architectural rather than ontological. Measurement becomes intelligible when the operations hidden inside a reported value are separated: distinction, reference, increment, order, enumeration, recurrence, comparator, unit, scale, numeral representation, and bridge. Spatial and temporal descriptions participate jointly in this architecture. Neither needs to be declared prior to the other.

Cyclic return then provides a unifying calibration. The same phase can close while history advances. Frequency counts such returns against a temporal comparator. Period reads the reciprocal orientation. The SI second supplies a physical standard by fixing a reproducible atomic frequency. Pitch systems add further cyclic and multiplicative registries on top of that temporal unit. A Hertz label therefore records much more than a bare number.

Numeral base enters at the representation layer but has mathematically consequential effects on visibility. Base twelve does not create the prime factors 2 and 3, yet it makes their divisibility terminate naturally in positional subdivision. A duodecimal multiplication table displays that structure through an actual twelve-digit alphabet and a zero-bounded Cartesian grid. The difference between triangular accumulation and square product enumeration further shows that even familiar counting operations depend upon the geometry of the registry used to externalize them.

The inscribed/circumscribed construction supplies a geometric analogue of the same discipline. One cyclic whole supports two finite witnesses. Their reciprocal relation records orientation of comparison; their narrowing bracket records finite resolution; and their powers across dimensional grade preserve a generating scale relation. At triangular minimum closure the exact linear 1:2 relation and area 1:4 relation no longer look like competing answers. They become different-grade expressions of one construction.

Pythagorean tuning supplies the complementary arithmetic case. Exact prime ratios generate an indefinitely nonclosing octave quotient, while a twelve-class projection and twelve-tone equal-tempered system impose different finite closure rules. The Pythagorean comma then becomes a paradigmatic residue: not noise, not failure of arithmetic, but the exact remainder left when two independently coherent routes do not reattach under the intended identification.

The relativistic closure bridge completes the inquiry that originally motivated the geometric portion of the report. Coarsening a finite n-gon truncates the continuous curve description until the triangle supplies the last ordinary polygonal enclosure. After endpoint normalization, that bounded geometric relation can carry the independently established Lorentz factor without changing the physical law. Spatial contraction, reciprocal spatial dilation, proper-time slowing, and coordinate-time dilation can therefore be displayed as oriented readings of one transported factor, while accelerator motion and atomic-clock differentials remain tied to their own declared physical models. The value of the construction lies precisely in keeping those layers connected without declaring them identical.

INTERPRETIVE - Condensed methodological statement

Define the relation before naming its value. Preserve the marker that makes a distinction. Record the operation that generates the increment. Keep phase return separate from accumulated history. Declare the unit and the numeral base. Treat reciprocal orientation and dimensional grade as information. Translate between registries only through an explicit bridge, and record whatever the bridge does not carry.

Appendix A. Duodecimal notation and multiplication registry

This paper uses the digit convention 0,1,2,3,4,5,6,7,8,9,A,B. The symbols A and B are single digits with decimal values ten and eleven. Rollover occurs at 10_12. The convention preserves zero as the positional placeholder and permits a genuine twelve-state one-place registry rather than a decimal list of the integers zero through eleven.

duodecimal	decimal translation
0_12	0_10
9_12	9_10
A_12	10_10
B_12	11_10
10_12	12_10
11_12	13_10
1A_12	22_10
1B_12	23_10
20_12	24_10
100_12	144_10
308_12	440_10
840_12	1200_10
1946716076_12	9,192,631,770_10

The full multiplication grid appears in Figure 5. Because its entries remain in base twelve, products such as B x B display as A1_12 rather than 121_10. This is the behavior needed when the goal is to inspect the divisibility and carry structure of the base itself.

Appendix B. Exact proofs and calibration notes

B.1 Equilateral triangle inscribed/circumscribed ratio without trigonometry

In an equilateral triangle, each median is also an altitude and angle bisector, and the centroid divides each median in the ratio 2:1 measured from the vertex. Because the circumcenter, incenter, centroid, and orthocenter coincide, the circumradius is two thirds of the altitude and the inradius is one third. Hence R = 2r for every equilateral triangle.

Now let one circle serve as the circumcircle of an inner equilateral triangle and as the incircle of an outer equilateral triangle. The inner triangle has circumradius equal to the circle radius. The outer triangle has inradius equal to the same circle radius. Since R=2r within an equilateral triangle, the outer triangle is similar to the inner triangle with linear scale factor 2. Perimeters therefore have ratio 1:2 and areas ratio 1:4.

$$P_{inner} : P_{outer} = 1:2 \qquad \text{and} \qquad A_{inner} : A_{outer} = 1:4$$

B.2 Base-b finite expansion theorem

Suppose a/c is reduced. If it terminates after m base-b places, it equals N/b^m for an integer N. Cross multiplication gives a b^m = N c. Because gcd(a,c)=1, c divides b^m. Therefore every prime factor of c divides b. Conversely, if c divides b^m, write b^m = c q. Then a/c = aq/b^m, which terminates after at most m places. This proves the criterion used in Section 11.

B.3 Exact Pythagorean nonclosure

Assume for contradiction that m>0 exact fifths equal n octaves: $(3/2)^m = 2^n$. Then $3^m = 2^{(m+n)}$. The left side contains only prime factor 3 and the right side only prime factor 2, contradicting unique prime factorization. Therefore the exact Pythagorean fifth generator has no nonzero finite octave return.

B.4 Conventional analytic chart for the polygon ratio

The primary construction in Section 14 defines the inner and outer polygonal witnesses directly from chord and tangent operations. For readers who want the conventional analytic chart, the regular n-gon perimeter ratio can be written as $P_n^-/P_n^+ = \cos(\pi/n)$. This identity is included only as a secondary coordinate verification; it is not used to define the registry, the full turn, or the triangle endpoint. The constructive definition remains the ratio of the measured inner chord perimeter to the measured outer tangent perimeter generated from the same n markers.

B.5 Decimal cents as a secondary translation

For conventional comparison with modern tuning literature, an interval ratio x can be expressed in cents by $c = 1200 \log_2(x)$. Under this chart, the exact Pythagorean comma 531441/524288 is approximately 23.460010 decimal cents. This decimal/logarithmic value is not used as the defining relation; the exact integer ratio remains primary.

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NOTICE - End note

This report intentionally preserves exact ratios, integer factorizations, declared turn fractions, and construction histories before optional analytic or decimal translation. Standard decimal values appear where interoperability with SI metrology and common musical practice requires them; duodecimal translations are supplied where base-sensitive structure is part of the question.